Amritha Mallikarjun

Email: amritham@upenn.edu
Mobile: 610-209-6937
U.S. Citizen

Background

Behavioral and data scientist with a Ph.D. in Neuroscience and Cognitive Science, specializing in the design and evaluation of learned systems under real-world constraints. Experienced in statistical modeling, machine learning, and large-scale longitudinal data analysis, with a track record of translating complex empirical findings for technical, policy, and public audiences.

Education

- **University of Maryland, College Park** — College Park, MD
  Ph.D. in Neuroscience and Cognitive Science — *2014 – 2020*
  M.S. in Neuroscience and Cognitive Science — *2014 – 2017*

- **Carnegie Mellon University** — Pittsburgh, PA
  B.S. in Cognitive Science — *2010 – 2014*
  B.S. in Human-Computer Interaction — *2010 – 2014*

Skills Summary

- **Programming**: Python, R, Java, LaTeX, Qualtrics, relational database design (Airtable), data integration (Zapier)
- **Statistical Methods**: Hierarchical models, linear and logistic regression, classification models, Bayesian analysis, supervised machine learning (XGBoost, gradient boosting)
- **Data Skills**: Management and analysis of large, longitudinal data sets, model evaluation
- **Data Visualization**: Statistical graphics in R (ggplot2), clear visual communication of complex data
- **Research Methods**: Experimental design, survey design, grant writing, technical and scientific writing
- **Communication**: Translating technical results for diverse audiences, multi- and inter-disciplinary collaboration
- **Additional Languages**: Tamil (native comprehension), French (working proficiency, 8 years of formal instruction)
- **AI Tools**: Active user of Claude and ChatGPT in research workflows; integrated AI tools into Airtable-based data systems for longitudinal study coordination and management

Experience

- **Senior Research Investigator & Data Scientist (formerly Postdoctoral Researcher)** — 2020–present
  Penn Vet Working Dog Center, University of Pennsylvania
  - **Study Development and Execution**: Design and conduct experimental, quasi-experimental, and longitudinal studies examining learning, decision-making, and performance under real-world constraints; translate broad research questions into clearly specified hypotheses, measurement strategies, and analytic plans
  - **Statistical Leadership and Model Specification**: Serve as primary quantitative lead across multidisciplinary projects; specify and compare hierarchical/mixed-effects, logistic, regression, and classification models; conduct robustness checks and diagnostic evaluation to ensure appropriate model assumptions and interpretable results
  - **Large-Scale Longitudinal Data Analysis**: Analyze complex behavioral and observational datasets ranging from 100 to 100,000+ observations; perform preprocessing, feature construction, model comparison, and evaluation; test alternative specifications to assess competing interpretations within multi-year datasets
  - **Research Funding and Communication**: Co-authored 9 funded grants (>$850K total) and 20+ peer-reviewed publications; communicate statistical findings through technical writing, data visualization (R/ggplot2), and presentations to academic, policy, and public audiences
  - **Predictive Modeling and Evaluation**: Built and evaluated machine learning models linking behavioral, physiological, and disease-detection data to outcomes; assessed model fit, generalization, and stability under noisy real-world conditions

- **Graduate Research Assistant** — 2014–2020
  University of Maryland, College Park
  - **Human Subjects Research**: Designed and conducted adult and infant behavioral studies examining cognitive control (flanker, task-switching, working memory), psycholinguistic processing (past tense generation, phoneme restoration), and speech perception in noise across monolinguals, second-language-learners, and clinical populations (ADHD); managed IRB protocols, participant recruitment, and data collection
  - **Comparative Cognition Research Program**: Developed an independent comparative research program examining speech perception, auditory category learning, and language acquisition across humans and domestic dogs; recruited and coordinated 700+ dog participants across 10+ studies using adapted infant behavioral paradigms (Headturn Preference Procedure)

- **Research Intern** — Spring 2018
  Rosetta Stone
  - **Behavioral Data Analysis**: Analyzed learner behavioral data to evaluate software effectiveness; conducted competitive analysis of speech recognition performance under varying acoustic conditions

Selected Applied Projects

- **ERP Investigation of Argument Structure Priming in Sentence Processing** 2014–2015
  - Designed and conducted EEG study examining P600 and N400 modulation under masked argument structure priming, investigating whether verb subcategorization bias influences syntactic reanalysis in garden-path sentences
  - Responsible for data analysis: EEG preprocessing in EEGlab including epoching, artifact rejection, and filtering; conducted ANOVA and linear regression on ERP amplitudes across time windows and electrode sites
  - Developed stimulus set of 270 sentences with normed verb subcategorization biases across three prime conditions
- **Preventing Military Working Dog Heat Injury** 2025–2026
  - Conducted literature review on heat injury prevention strategies for military working dogs as part of Army SBIR Phase I subcontract (Raekor/Penn Vet Working Dog Center)
  - Generated use case report evaluating novel cooling technology for application in austere field conditions
  - Designed Phase II efficacy study and budget including sample size calculations and power analysis
- **Exercise Enrichment and Scent Detection Performance (Homeland Security)** 2022–Present
  - Contributing technical researcher on DHS-funded project examining how exercise enrichment impacts scent detection performance in TSA detection dogs (Sub to Auburn University: DHSST-Contract 70RSAT22CB0000002, $989,214)
  - Building participant data collection infrastructure including automated reminder systems and web-based data collection tools, as well as database setup and workflow automation
  - Responsibilities include data cleaning, statistical modeling, and analysis
- **Olfactometer Software Redesign and Data Pipeline Engineering** 2022–Present
  - Led iterative redesign of research-grade control software for portable automated olfactometers, transforming a fragile codebase meant for expert researchers into a robust tool usable by non-technical personnel in practical and experimental settings
  - Rewrote validation logic and backend safeguards to prevent crashes from common edge cases; implemented user-friendly features such as defensive input handling, mid-session configuration changes, and trial-skipping functionality
  - Designed streamlined data packaging and automated upload workflows (Airtable integration) to improve reliability, reduce missing data, and standardize outputs for downstream statistical analysis
- **Environmental Monitoring: Chronic Wasting Disease (CWD)** 2021–Present
  - Led development and evaluation of non-invasive CWD surveillance methods using volatile organic compound odor signatures, combining detection dogs, longitudinal field data, and statistical modeling to inform wildlife disease monitoring
- **Speech Perception under Real-World Constraints** 2018–Present
  - Designed comparative studies of perception under noise and signal degradation to evaluate how attention and prior experience shape recognition performance in humans and domestic dogs

Selected Funding

For grants where I served as Co-Investigator, I developed and authored proposed statistical methods, analytic plans, and evaluation frameworks.

- **A Combined Sensor Approach to Examine Volatile Organic Compounds as Potential Markers of Canine Hemangiosarcoma (2026)**: Co-Investigator. Morris Animal Foundation ($211,590). Co-authored grant to examine the volatile organic compound signature of hemangiosarcoma, a deadly canine cancer, with detection canines and analytic methods.
- **Preventing Military Working Dog Heat Injury (2025–2026)**: Subcontractor. Army SBIR Phase I (Raekor; $29,500). Contributed to Phase I and Phase II proposal and budget development; conducted literature review, generated use case report, and designed Phase II efficacy study with power analysis.
- **Canimetrics Learning Platform and Data Infrastructure (2026)**: Co-Investigator. American German Shepherd Foundation ($10,000). Developed statistical and data architecture plan for an online canine fitness learning platform and multi-site data collection system.
- **Wild CWD Surveillance via VOC Identification (2024–2025)**: Co-Investigator. USDA ($243,578). Developed statistical strategy for identifying volatile organic compound signatures differentiating CWD-positive and negative deer; supported training aid validation.
- **Monitoring Chronic Wasting Disease with Bio-Samples (2023–2024)**: Co-Investigator. USDA ($249,924). Developed evaluation framework for detection-dog candidate selection and diagnostic validation.
- **Visual Scale of Heat-Related Illness (2022–2023)**: Co-Investigator. Association for Professional Dog Trainers ($2,471). Designed statistical validation approach for perceptual rating accuracy.

Press and Media

- **Broadcast Media**: Featured in PBS NOVA documentary *Can Dogs Talk?* (2024), Netflix documentary *Inside the Mind of a Dog* (2024) and CBS *Mission Unstoppable* (2021)
- **Major Media Coverage**: Research covered by *Scientific American*, *National Geographic*, *Smithsonian Magazine*, *The Washington Post*, *Popular Science*, *Newsweek*, and ABC News
- **Public Communications**: Presented research through public talks and academic conferences and provided expert commentary on newly published research in national newspaper and magazine coverage

Selected Publications

Full list available upon request. For all papers listed below, I developed and authored statistical methods and results.

- **Mallikarjun, A.**, Oslin, D. W., & Otto, C. M. Biological sensor detection of volatile organic compounds associated with post-traumatic stress disorder in the blood plasma of major depressive disorder patients. [veteran population] Under review at the *Journal of Veterinary Behavior*.
- Powers, G. J., Wilson, C., Reetz, J. A., **Mallikarjun, A.**, Capparell, T., and Otto, C. M. (in press). Thoracic spondylosis deformans in search-and-rescue dogs. *American Journal of Veterinary Research.* [September 11, 2001 disaster response cohort]
- **Mallikarjun, A.**, Shroads, E., & Newman, R. S. (2024). Perception of vocoded speech in domestic dogs. *Animal Cognition*, 27(1), 1–13.
- **Mallikarjun, A.**, and Newman, R. S. Effect of number of background talkers on speech perception in the domestic dog. Under review at *Applied Animal Behaviour Science*.
- **Mallikarjun, A.**, Shroads, E., & Newman, R. S. (2019). The cocktail party effect in the domestic dog (*Canis familiaris*). *Animal Cognition*, 22(3), 423–432.
- **Mallikarjun, A.**, Shroads, E., & Newman, R. S. (2021). The role of linguistic experience in the development of the consonant bias. *Animal Cognition*, 24(3), 419–431.
- **Mallikarjun, A.**, Shroads, E., & Newman, R. S. (2023). Language preference in the domestic dog. *Animal Cognition*, 26(2), 451–463.
- Tsui, A. S. M., Erickson, L. C., **Mallikarjun, A.**, Thiessen, E. D., and Fennell, C. T. (2021). Dual language statistical word segmentation in infancy: Simulating a language-mixing bilingual environment. *Developmental Science*, 24(3), e13050. [human infant participants; study concept originated from undergraduate honors project]
- **Mallikarjun, A.**, Charendoff, I., Moore, M. B., Wilson, C., Nguyen, E., . . . , & Otto, C. M. (2025). Dogs can generalize from cotton training aids to fecal matter in chronic wasting disease detection. *Journal of Wildlife Diseases*, 61(3), 600–608.
- **Mallikarjun, A.**, Swartz, B., Kane, S. A., Gibison, M., Wilson, I., . . . , & Otto, C. M. (2023). Canine detection of chronic wasting disease (CWD) in laboratory and field settings. *Prion*, 17(1), 16–28.
- Kane, S. A., Lee, Y. E., Essler, J. L., **Mallikarjun, A.**, Preti, G., Verta, A., . . . , & Otto, C. M. (2022). Canine discrimination of ovarian cancer through volatile organic compounds. *Talanta*, 250, 123729.
- Gokool, V. A., Crespo-Cajigas, J., **Mallikarjun, A.**, Collins, A., Kane, S. A., Plymouth, V., . . . , & Otto, C. M. (2022). The use of biological sensors and instrumental analysis to discriminate COVID-19 odor signatures. *Biosensors*, 12(11), 1003–1028.

Teaching and Mentorship

- **Teaching, Curriculum Design, and Instruction**: Designed and taught annual summer seminar series for 8–10 undergraduate research interns at Penn Vet (2021–2025) on academic writing, research methods, and professional development; served as lab instructor for experimental design course at Penn Vet School of Veterinary Medicine (2024–2025); delivered 8+ guest lectures at UPenn, UMD, and Barnard College; tutored high school students in intermediate and advanced French, including designing curriculum for advanced learners (CMU, 2013–2014)
- **Student Mentorship**: Mentored 10+ students (undergraduate, post-baccalaureate, master's, veterinary, and high school) through full research cycles including experimental design, data collection, statistical analysis, and manuscript preparation; 5 mentored projects resulted in peer-reviewed publications